

Did hydration breaks change the 2026 World Cup?

A reproducible analysis of 203 mandatory breaks across 102 matches

Version 1.0 · 2026-07-25 · github.com/rezaba002/WC2026-HydrationBreakAnalysis

Every figure in this report is generated by a script from audited source data. The analysis specification was frozen before any outcome was computed; every subsequent amendment and limitation is dated and justified in `CHANGELOG.md` (A1–A5).

1. The question

The 2026 World Cup was the first to mandate hydration breaks in **every match regardless of weather** — roughly three minutes near the 22nd minute of each half. Previously, cooling breaks were triggered only by a heat threshold. Football was, in effect, divided into four quarters.

The breaks were immediately controversial. Managers said they changed the identity of the game. Commentators narrated momentum swings around them. Broadcasters sold them.

This report asks one question:

Did the universal hydration-break policy measurably change attacking performance — and if the average effect was small, why were the breaks so widely experienced as decisive?

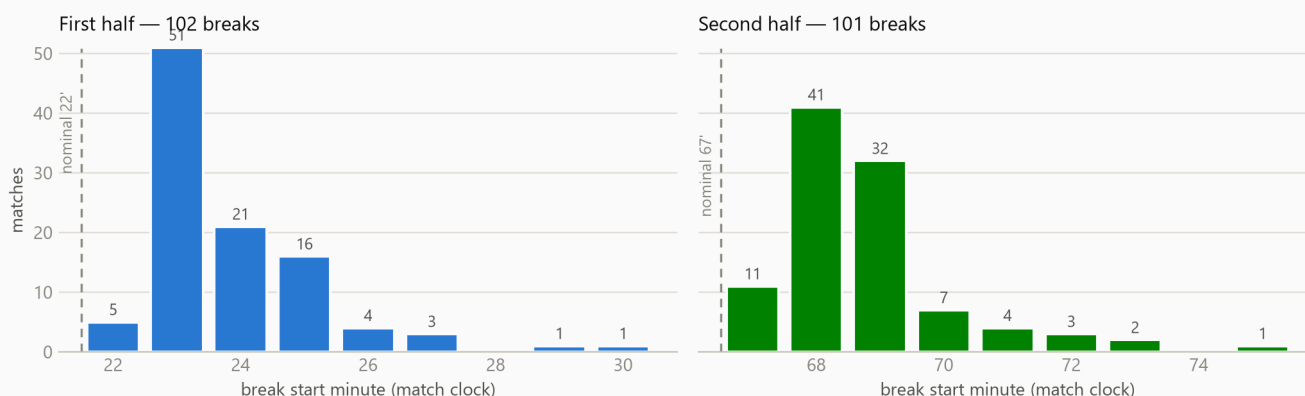
The interesting possibility is not that one side of that is wrong. It is that both are true at once: a near-zero average effect coexisting with real, measurable changes to how football was coached, staffed, timed and narrated.

2. What the breaks actually looked like

Before testing anything, the basic facts.

Hydration breaks started later than the nominal minute

Actual start minutes of the 203 recorded breaks, WC2026 (102 matches). Modal starts: 23' and 68'.

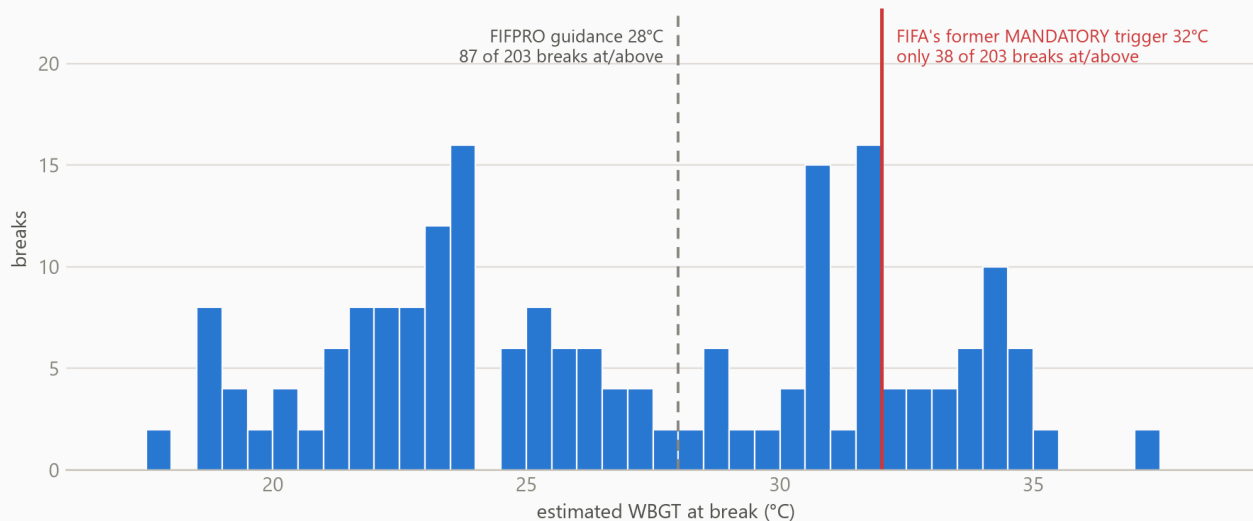


Breaks did not happen at the nominal minute. They clustered at **23'** and **68'**, with a tail out to 30' and 75' — referees waited for a natural stoppage. Of 203 recorded breaks, 102 were first-half and 101 second-half. (One half, France–Iraq's second, has no recorded break in any source we hold; it is documented, not silently dropped.)

This matters for measurement: an analysis anchored to 22'/67' would misplace most breaks. All results here use each match's **actual** break minute.

Four in five breaks happened below FIFA's own former mandatory threshold

Estimated wet-bulb globe temperature at each of the 203 breaks. Median 26.1°C. FIFA and FIFPRO thresholds differ — see report §2.



The heat context explains why the policy was contested. Estimated wet-bulb globe temperature at the breaks ranged from 17.8 °C to 37.2 °C, median 26.1 °C.

Two different thresholds matter here, and they belong to two different bodies [3][4]. FIFA's own protocol made cooling breaks **mandatory at WBGT ≥ 32 °C**, and merely requestable between roughly 27 °C and 32 °C. FIFPRO, the players' union, recommends breaks from about 26 °C and argues for delaying matches above 28 °C. Against those benchmarks:

threshold	breaks at or above	share
FIFPRO break guidance, 26 °C	103	51%
FIFPRO delay guidance, 28 °C	87	43%
FIFA former mandatory trigger, 32 °C	38	19%

So **four out of five 2026 breaks happened in conditions where FIFA's own previous rules would not have required one at all**. This is a policy change, not a weather event — which is precisely why its effects are worth measuring.

3. How the effect was measured

Two design decisions carry the whole analysis.

The clock problem. The match clock runs during a break. A naive "10 minutes before versus 10 minutes after" comparison therefore gives the *after* side about three fewer minutes of actual football. Every result below is computed on two clocks: the **display clock** (what viewers and event feeds see) and the **break-adjusted clock** (hydration dead time removed — and only that; throw-ins and VAR remain, so it is not an "active play" clock and is never described as one).

The counterfactual problem. Comparing before-and-after tells you nothing on its own, because football is volatile: dominance decays, pressure reverts to the mean. The test must be *versus what normally happens*. For each real break we enumerate eligible non-break minutes in the same match and half, matched on score state and stage, excluding minutes near goals, red cards, VAR reviews and half boundaries — then draw 10,000 randomized pseudo-breaks and compare the real result against that null distribution.

Primary outcome is **balance disruption**: the absolute change in shot differential across the break, $|\Delta(\text{home} - \text{away})|$. It is direction-agnostic, so it can be scored for every break, and it is always defined — unlike shot *share*, which is 0/0 in the many quiet windows. Uncertainty is a match-level bootstrap (5,000 iterations, fixed seed), because two breaks per match are not two independent observations.

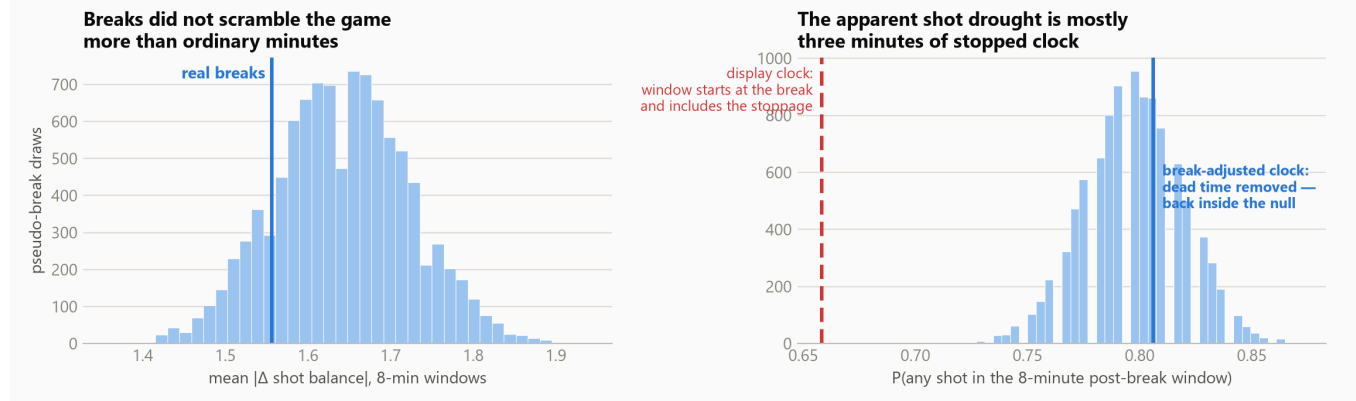
A note on what is not here. Per-shot xG does not exist in any independently auditable source for this tournament. FIFA's official post-match reports publish shot logs without xG; FBref blocks automated access. Outcomes are therefore shot-based (CHANGELOG A2). This is stated rather than papered over.

4. The seductive result, and its correction

Real hydration breaks vs 10,000 moments where nobody stopped the game

196 breaks; pseudo-breaks matched on half, score state and stage.

Post-break window measured two ways — display clock (starts at the break, includes the ~3-min stoppage) vs break-adjusted clock (dead time removed).



The outcome is the probability of any shot in the **8-minute post-break window** — and the two clocks define that window differently, which is the whole point:

- **Display clock:** the window begins when the break is called, so it *contains* the ~3-minute stoppage. This is what a viewer or an event feed sees.
- **Break-adjusted clock:** the hydration dead time is removed, so the window holds 8 minutes of displayed football.

On the display clock, the breaks look devastating. The probability of a shot in that window falls to **0.658**, against a null of ~0.80 — far outside the range of ordinary matched minutes. If you stop there, you have a headline: hydration breaks kill momentum.

Remove the three dead minutes and it vanishes. On the break-adjusted clock the same probability is **0.806** against a null of **0.822** — squarely inside the null distribution. The apparent post-break shot drought was largely the stopped clock being counted as football. ("Momentum" in the broader sense — possession, territory, tactical control, psychology — is not what this measures; shots are.)

The primary metric agrees. The cleanest way to state it is the **paired effect**: for each break, its observed disruption minus the mean disruption across that break's *own* matched control minutes, averaged and bootstrapped by match.

Effect: -0.085 shots, 95% match-clustered CI [-0.238, +0.074] (196 breaks, 102 matches).

That interval includes zero. The point estimate is negative — real breaks were followed by marginally less disruption than comparable ordinary minutes — but the difference is not distinguishable from no difference once uncertainty is placed around the contrast itself and clustered by match. This is deliberately more conservative than the randomization percentile, which reflects only the variability of the controls, not the match-to-match variability of the real breaks.

Note what the interval still excludes: even its break-unfavourable end is **+0.07 shots** of extra disruption — orders of magnitude smaller than the decisive swings described publicly. The finding is *no detectable*

difference, not breaks calmed the game.

The estimate is stable when weakly-matched breaks are dropped (≥ 3 , ≥ 5 , ≥ 10 control minutes per break), so thin matching is not driving it.

The average competitive effect is null. This replicates the existing preprint (arXiv 2607.19783) by an independent route, and the older momentum-index audit agrees: treated breaks showed a mean momentum drop of 7.91 versus 7.11 at historical control minutes — a difference of +0.80 with a 95% interval of [−2.11, +3.85], straddling zero.

A result we deleted

The analysis initially produced a strong directional finding: home teams gained shot differential after breaks (+0.29 after placement matching, with essentially no null draws reaching the observed value). It survived every robustness test applied to it.

It is not reported as a finding, because the control pool itself is biased for directional contrasts. All unfiltered eligible minutes average **+0.040** — essentially zero, as symmetry demands. The score-state-matched control pool averages **−0.134**, and is negative in *all three* score-state buckets. That is impossible for a clean control: a baseline tracking game state would be negative when the home side leads and sits back, and **positive** when it trails and chases. The cause is the preregistered goal-proximity screen, which strips disproportionately many home-pressure phases out of the controls while real breaks sample them at the natural rate.

The headline is unaffected — the primary metric is absolute, and the same bias runs *conservative* there. Full evidence: `reports/tables/robustness.md` ; limitation logged as CHANGELOG A5.

Does the null hold up?

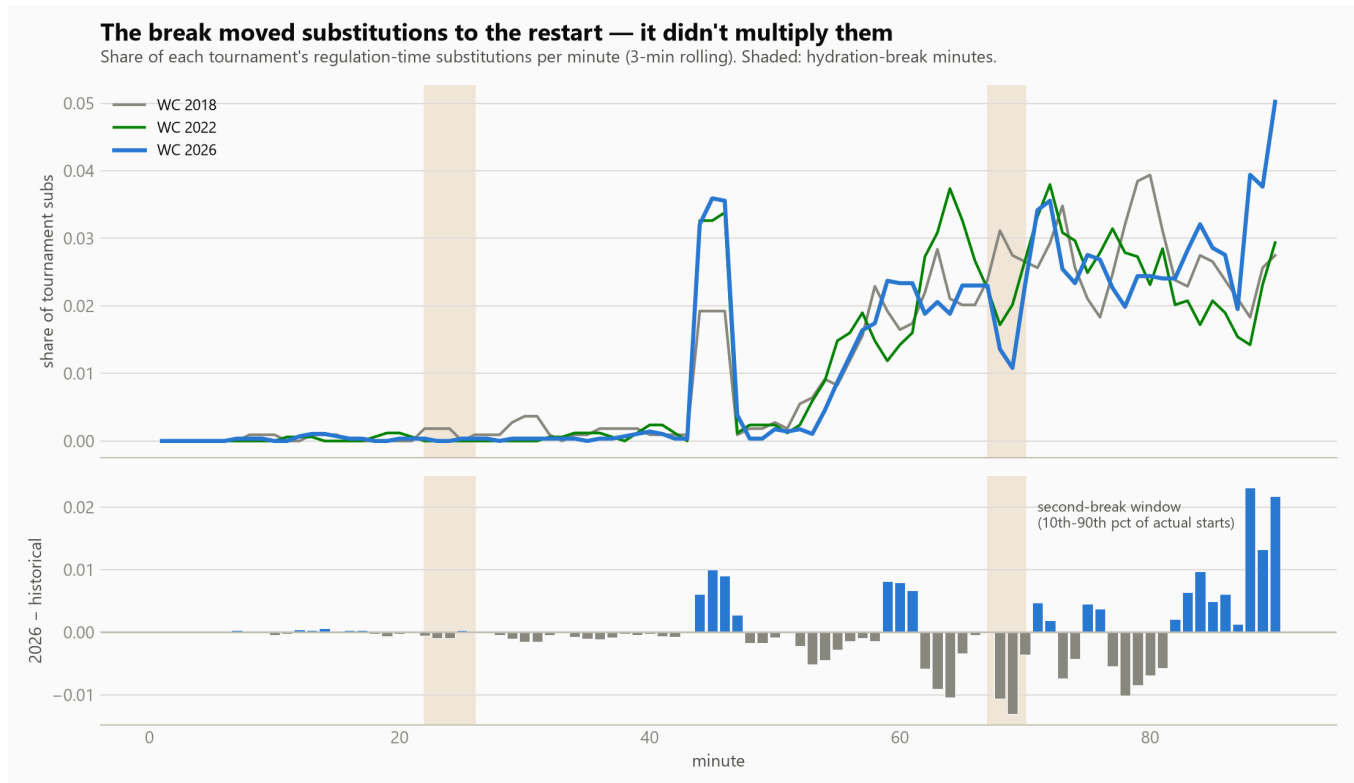
check	result
Paired effect, ≥ 3 / ≥ 5 / ≥ 10 control minutes	−0.081 / −0.071 / −0.170 — estimate and interval stable
Leave-one-match-out (102 refits)	gap stable in [−0.224, −0.158]; no single match drives it
Drop windows containing a red card	1.456 vs 1.649 — unchanged
Drop breaks with a goal in the prior 3'	1.442 vs 1.664 — unchanged
Nominal 22'/67' timing instead of actual	1.549 vs 1.608 — unchanged
Group vs knockout stage	1.458 vs 1.711 · 1.451 vs 1.520 — both below null
Windows of 5 / 8 / 10 minutes	consistent throughout

One exploratory exception, flagged and not claimed: the halves diverge. First-half breaks are markedly *less* disruptive than matched minutes (1.10 vs 1.59), while **second-half breaks are the one subgroup sitting above its null** (1.87 vs 1.72). That is coherent with the substitution evidence below — but it is one subgroup cut among several and needs preregistration before it can be a claim.

5. What did change

The competitive null is only half the story. Several things about World Cup football did measurably change.

Substitutions moved — they did not multiply



The intuitive test fails. Only **18.7%** of 2026 second-half substitutions fell within ± 3 minutes of their own match's break — *below* the minute-matched historical expectation (19.6% from 2018, 20.7% from 2022). The break did not become a substitution magnet.

The displacement pattern underneath it is the real finding (exploratory, not preregistered): a **deficit in the three minutes before the stoppage** (6.8% vs 7.6/8.5% historically) and a **surplus in the three minutes after the restart** (15.0% vs 12.6/14.6%). Coaches did not make more changes; they waited for the break and made them at the restart. The break became a scheduling point.

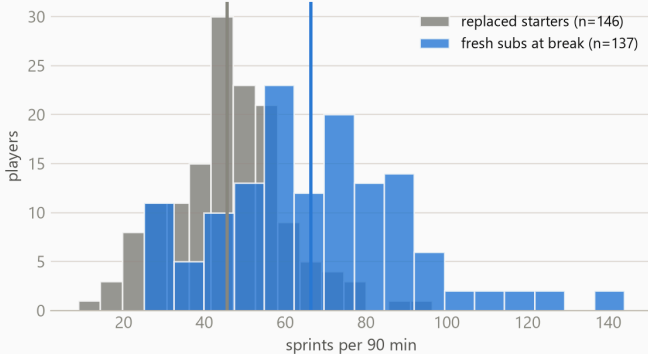
Context: 2026 averaged 4.64 regulation substitutions per team-match against 4.39 in 2022 and 2.87 in 2018 — the jump is the five-substitution rule, not the breaks. Median first substitution: 58' (2026), 57' (2022), 61' (2018).

The break as a fresh-power window

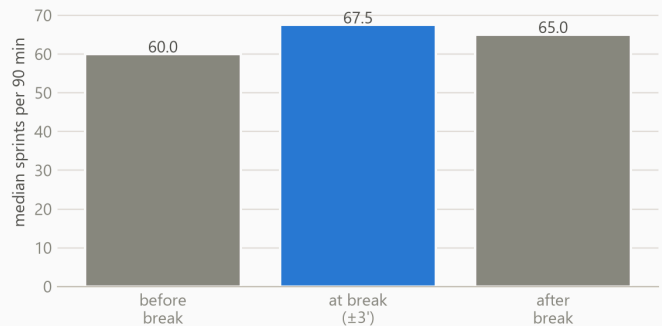
The second break as a fresh-power window

Per-90 physical output of WC2026 substitutes (FIFA match totals). Deployment, not physiology.

Fresh legs sprint far more than the legs they replace



Entry timing barely matters (matched appearance length)



FIFA's per-player physical data allows one legitimate question — not whether breaks refreshed players (that is unmeasurable here), but **whether coaches used them to inject fresh legs**.

Substitutes entering at the break ran at **66.3 sprints per 90** against **45.7** for the starters they replaced; high-speed running 162 vs 121, high-speed distance 1,245 m vs 806 m. Fresh legs are dramatically more explosive than tiring ones — as expected.

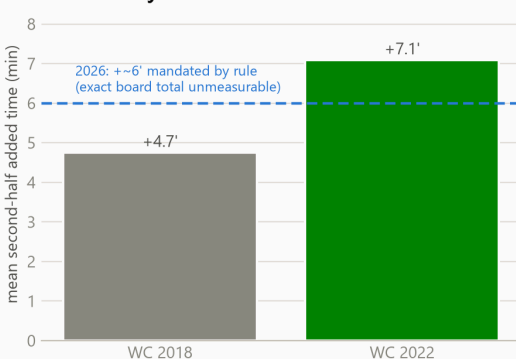
Honesty check: entry *timing* barely matters. Restricted to a matched ≤ 30 -minute appearance band, median sprints/90 are 60.0 before the break, 67.5 at it, 65.0 after — the timing differences largely collapse. The fresh-versus-tired gap is the story; the break's role is that it is where the swap gets coordinated.

Matches got longer — but 2022 got there first

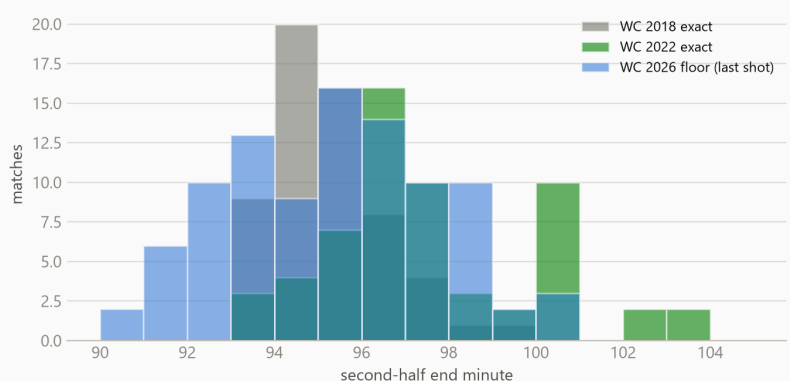
Did hydration breaks make matches even longer?

2018/2022 exact half-ends (StatsBomb); 2026 is a last-shot lower bound (exact board minutes unavailable).

2022 already stretched the clock



How late the second half ran



Qatar 2022 was the added-time anomaly: second halves ran to **97.1'** on average (+7.1), against **94.7'** (+4.7) in 2018. 2026 adds roughly six minutes per match **by rule**, since the breaks sit inside the clock.

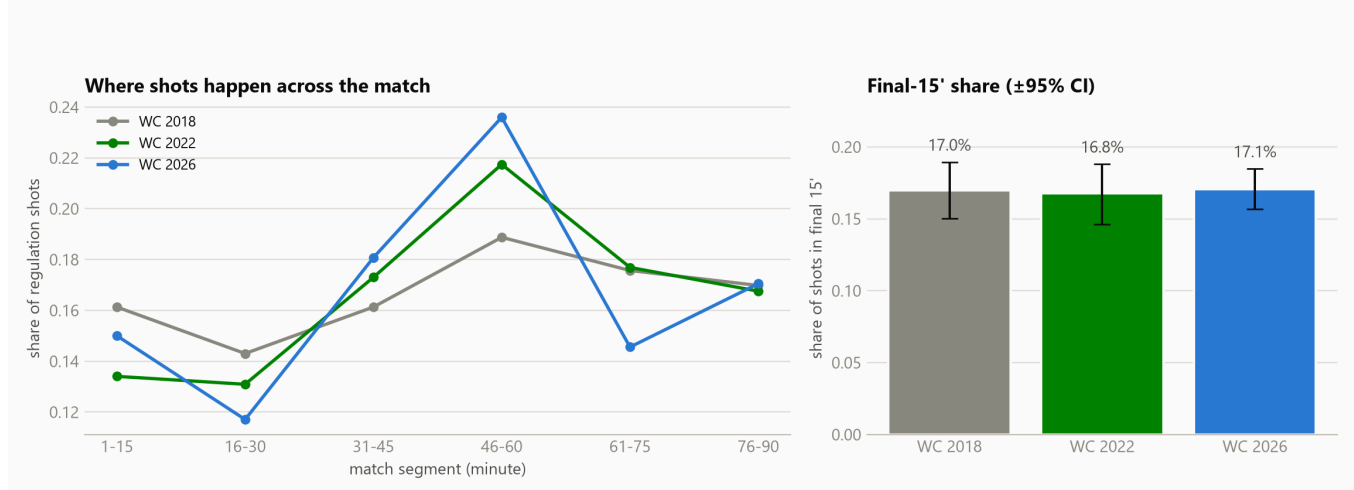
Whether 2026's boards *exceeded* the already-inflated 2022 environment cannot be determined: exact added-time minutes are absent from every auditable source we hold (the FIFA feed strips stoppage notation in 97 of

104 matches). We report a last-shot lower bound and no more. Late scoring shows no clean trend either — goals at $\geq 90'$ were 11.4% (2018), 8.9% (2022), 12.6% (2026).

Did the game stay alive later?

Did the game stay alive later once breaks existed?

Behavioral proxy — not physical. 2026-vs-2022 isolates the break era (both 5-sub); 2022-vs-2018 is the sub-rule jump.



The physical-recovery question — did breaks help players sustain the game? — cannot be answered directly: no per-player physical data exists for 2018 or 2022, and 2026's data are match totals with no pre/post-break split. (This was searched and confirmed, not assumed.)

The available proxy is behavioral: the share of shots in the final 15 minutes. It is **flat across all three tournaments** — 17.0% (2018), 16.8% (2022), 17.1% (2026), with overlapping confidence intervals. Whatever the breaks did, they did not visibly change how alive the closing stages were.

6. Why it felt different

If the average effect is null, the perception needs explaining. This is where the project departs from prior work.

We preregistered an inclusion rule — a claim counts only when a commentator, manager, player or major outlet attributes a change in a **specific match** to a **specific break** — then collected in two passes:

1. a **topical pass** across outlets writing about the breaks (Opta Analyst, Goal.com, Middle East Eye, The Times, TNT Sports, ESPN, England Football);
2. a **pre-specified stratified random sweep** of 24 matches / 48 breaks, each searched with an identical query template, logged in `data/manual/perception_sweep_log.csv` including every match where nothing was found.

The sweep is what makes the denominator credible. Searching only the matches *likely* to have claims would have rebuilt the exact bias under measurement, so the sample was fixed before any searching and every drawn match was searched whatever the result.

That yielded **22 usable claims and 24 logged rejections**. Every claim was then re-read against its source URL. Nineteen were confirmed at source; **three were not** and are excluded from all figures — one podcast that could never be located, one page that blocks automated access, and one quote absent from the page it was cited to. Two surviving claims needed correction: a break number (the source specified the second-half break, not the first) and a claim text (the source does not say the goal came a minute after the break; our own timeline puts them thirteen minutes apart). Both failures and both corrections originated in search-engine summaries that did not survive contact with the actual page — which is why the verification pass exists.

Each verified claim was then evaluated **blind to its text**: only the match, the break and the team said to benefit were read, and the swing was scored against that same match-half's own pseudo-break null. Several breaks drew claims from more than one outlet, so the statistically independent measure deduplicates to unique claimed breaks:

Seven of thirteen unique claimed breaks were supported (54%) — claim-level, for comparison, is 8 of 17 (47%). The two differ because heavily covered incidents contribute several rows; here that pulls the claim-level rate *down*, since the most-covered break (England–DR Congo's first, three separate claims) is not supported. Either way: the people making these claims were right about as often as not — and when right, often dramatically so.

The bias is not in the claims. It is in the **denominator**, and the random sweep measures it directly:

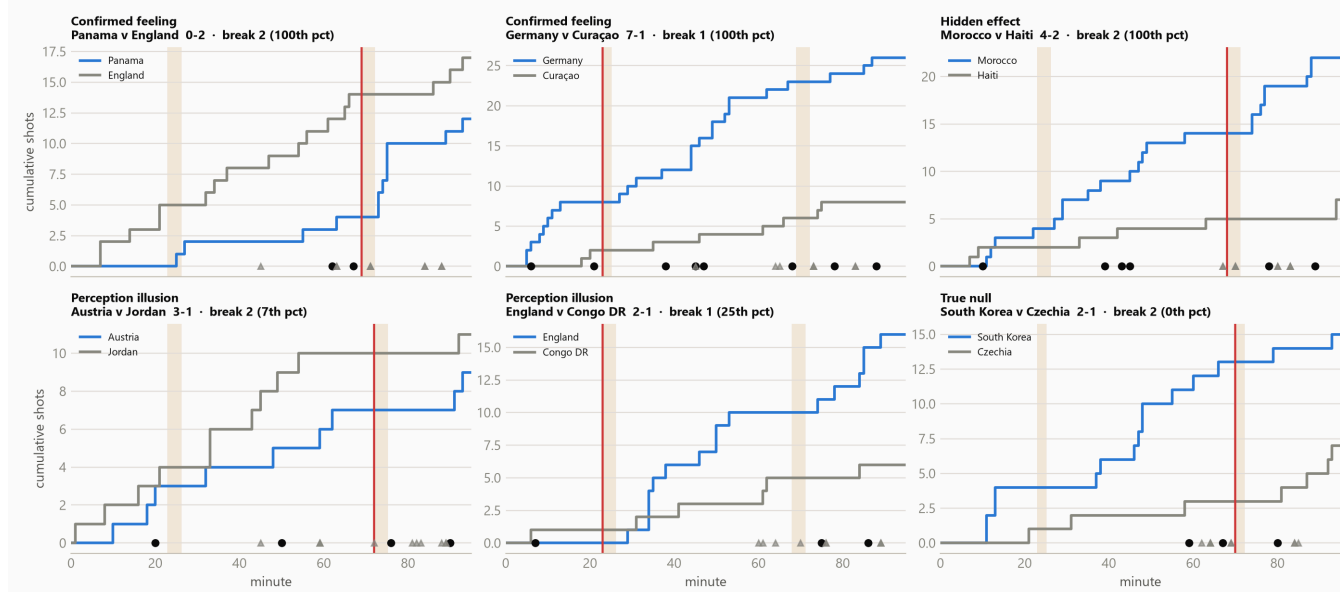
Public claims attached to just **4 of 48 randomly sampled breaks — 8.3%** (95% CI 3.3–19.6%), extrapolating to roughly **17 of 203 breaks** tournament-wide. Within that small, self-selected set the claims are right roughly half the time. Pundits were describing real swings. They were describing them from the tail of a distribution, and the tournament-wide story was then written from that tail.

The case-study matrix makes the asymmetry concrete. Scoring every break on swing size and crossing it with whether any claim was located:

	large swing	small swing
publicly claimed	3	4
no claim located	64	73

Six breaks, chosen by matrix — not by fame

Shaded = hydration breaks · red line = the break in question · dots = goals · triangles = substitutions on. Percentile is vs the same match-half's ordinary minutes.



Cases were selected by this matrix, not by fame. Germany–Curaçao qualified on merit at the 100th percentile. Netherlands–Sweden also scored at the 100th percentile but was displaced from the final six by Panama–England's larger swing; Switzerland–Bosnia, the third planning-stage favourite, never qualified at all. The hidden-effect slot went to Morocco–Haiti's second break — a 100th-percentile swing with no located commentary.

One case changed cells when the sweep ran, and it is worth reporting as such. Panama–England's second break is the tournament's largest pressure inversion: Panama went from 1 shot to England's 3 before it, to 6 against England's 0 after it — and still lost 0–2. The earlier topical pass found no commentary, so it was classified as a "hidden effect" and flagged *provisionally*, on the explicit ground that "no claim found" is not "no claim exists". The systematic sweep then located a TNT Sports data piece reporting precisely that dip. It is now a confirmed-feeling case. The caveat did its job, and the correction is left visible rather than quietly absorbed.

Meanwhile Austria–Jordan, which *did* generate public claims, sits at the **7th percentile** — the swing people remembered was smaller than a typical quiet passage of that same half.

Large post-break swings were common. Commentary about them was not. What determined whether a swing became a story was largely whether the scoreboard followed.

Coverage caveat: the sweep covers 24 of 104 matches. Its 8.3% rate is an unbiased estimate with a wide interval; `has_claim = False` for unswept matches means "no claim located", not "none exists".

7. Conclusion

Hydration breaks did not, on average, produce unusually large changes in shot balance. Once the stoppage itself was removed from the clock, post-break shooting activity looked ordinary. Measured against thousands of statistically comparable moments where nobody stopped the game, the football after a break

looks like ordinary football. The effect that seems obvious on a television clock is an artifact of counting three stopped minutes as play. This holds across windows, stages, exclusions, and every match left out in turn.

But the sport around the scoreboard did change. Two guaranteed tactical meetings per match became a fixed feature of coaching. Substitutions were re-timed to the restart — coordinated rather than multiplied. Matches carry roughly six mandated extra minutes on top of an already-stretched stoppage environment. And breaks became a scheduled window for injecting players who run half again as hard as the ones they replace.

And the perception was not a delusion — it was a sampling problem. When people made a specific, testable claim, the measured swing supported them roughly half the time, and sometimes very strongly. What never entered the conversation was the large majority of breaks that generated no public narrative at all — whether or not ordinary football volatility produced a swing afterwards. Our own case matrix contains dozens of breaks with swings as large as the famous ones and no located commentary. A rule applied 203 times was judged on the handful of occasions when the scoreboard happened to move immediately afterwards.

That is the honest shape of it:

The breaks did not consistently produce unusual swings in shot balance. They changed how coaches managed games, how long matches ran, how broadcasters packaged them — and they handed audiences a vivid, memorable, and deeply unrepresentative sample of evidence about themselves.

On the commercial question, the defensible line: **FIFA presented the breaks as a player-welfare measure; broadcasters discovered substantial new advertising inventory.** Both are documented. Inferring the second as the motive for the first is not supported by anything measured here, and this report does not claim it.

8. Limitations

Stated plainly, because they bound every claim above.

1. **No per-shot xG** in any auditable source; outcomes are shot-based.
2. **Directional results are not reportable** — the control pool is biased for signed contrasts (CHANGELOG A5). Only absolute metrics carry conclusions.
3. **2026 added time is unmeasurable** at board level; a lower bound is reported instead.
4. **Physiology is out of scope.** No claim here concerns player hydration, core temperature, or injury. Public event data cannot support them.
5. **Perception collection covers 24 of 104 matches** by random sweep (plus a topical pass): 22 claims, of which 19 were confirmed at source and 3 could not be. Only confirmed claims are used in any figure. One source (ESPN) blocks automated access and its claim awaits a manual browser check; two others should be treated as withdrawn unless a primary source is produced.

6. **Coverage is 102 of 104 matches, 203 of a possible 204 breaks.** Every exclusion is listed with a reason in `data/processed/exclusions.csv`.
7. **Substitution minutes in the 85–90' tail are not comparable across sources** — a recording artifact, flagged and not interpreted.
8. **Everything here is association**, not causation. Breaks were not randomly assigned; the placebo design asks whether post-break football was unusual, not whether the break caused it.

9. Reproducing this

```
pip install -r requirements.txt
python -m src.audit           # source inventory + hashes
python -m src.build_tables    # master tables
python -m src.placebo         # confirmatory placebo (Core Output 3)
python -m src.robustness      # robustness pass
python -m src.subs            # substitution timing (Core Output 4)
python -m src.physical        # fresh legs
python -m src.late_game       # late-game proxy
python -m src.added_time      # added time (Core Output 6)
python -m src.perception      # perception evaluation (Core Output 5)
python -m src.case_studies    # case selection (Core Output 7)
python -m pytest tests -q     # 29 tests
```

Sources, commits and SHA-256 hashes: `config/sources.yaml`, `data/processed/source_inventory.csv`.

Frozen specification: `spec.md`. All amendments: `CHANGELOG.md`.

10. References

Policy and rules

[1] FIFA (primary source). *Players to benefit from hydration breaks at FIFA World Cup 2026*. States that the referee stops play 22 minutes into each half, that each break lasts up to three minutes, that the time is **added back as stoppage time**, and that the breaks apply in all matches with no weather or temperature condition attached. <https://inside.fifa.com/organisation/news/hydration-breaks-world-cup-2026-player-welfare> Contemporary reporting and criticism: https://www.espn.com/soccer/story/_/id/48945011/why-there-drinks-breaks-2026-world-cup-fifa-criticised

[2] IFAB, *Laws of the Game*, Law 7 (The Duration of the Match). Distinguishes a **drinks break** (≤ 1 minute) from a **cooling break** (90 seconds to 3 minutes), and requires allowance for time lost. <https://www.theifab.com/laws/latest/the-duration-of-the-match/>

[3] FIFA heat protocol (pre-2026): cooling breaks mandatory at WBGT ≥ 32 °C, requestable roughly 27–32 °C. This project did not obtain FIFA's protocol document directly; the threshold is taken from secondary reporting and should be treated as such — <https://kestrelinstruments.com/blog/fifas-hydration-break-rule-explained-what-it-means-for-the-2026-world-cup> and the FIFA–FIFPRO threshold dispute reported at

https://www.espn.com/soccer/story/_/id/49343877/fifa-world-cup-fifpro-temperature-heat-protocol-talks-sources. Anyone citing the 32 °C figure in a publication should verify it against FIFA's own medical/competition regulations.

[4] FIFPRO heat-stress guidance: breaks recommended from ~26 °C WBGT; match delay advised above 28 °C. <https://fifpro.org/en/supporting-players/health-and-performance/extreme-weather-and-climate-change/> FIFA-FIFPRO threshold disagreement reported at https://www.espn.com/soccer/story/_/id/49343877/fifa-world-cup-fifpro-temperature-heat-protocol-talks-sources

Prior work

[5] *Do In-Match Hydration Breaks Alter Match Momentum? A Within-Match Case-Crossover Analysis of the 2026 FIFA World Cup*. arXiv:2607.19783. <https://arxiv.org/html/2607.19783>

[6] Opta Analyst, *Are World Cup Hydration Breaks Really Killing Momentum?* <https://theanalyst.com/articles/world-cup-2026-hydration-breaks-momentum-stats>

Data sources

[7] FIFA Post Match Summary Reports (PMSR), fifatrainingcentre.com — shot logs, substitutions, per-player physical data, team statistics. Retrieved 2026-07-24; per-file hashes in `data/processed/source_inventory.csv`.

[8] StatsBomb Open Data — World Cup 2018 and 2022 events (substitutions, shots, goals, period ends). <https://github.com/statsbomb/open-data>

[9] Ddey07/wc2026-hydration-momentum — break timings, treated/control covariates, SofaScore-derived momentum summaries. <https://github.com/Ddey07/wc2026-hydration-momentum>

[10] mominullptr/FIFA-World-Cup-2026-Dataset — tournament metadata, venues with coordinates, lineups. <https://github.com/mominullptr/FIFA-World-Cup-2026-Dataset>

[11] anup4khandelwal/wc26-analytics — PMSR parser approach (code reference only; no numbers used). <https://github.com/anup4khandelwal/wc26-analytics>

Perception sources

Full per-claim source URLs, verification status and rejection reasons: `data/manual/perception_claims.csv`, `perception_rejections.csv`, `perception_sweep_log.csv`. Outlets cited: Opta Analyst, Goal.com, Middle East Eye, The Times (secondhand), TNT Sports, England Football.

Commercial context

[12] Reporting on hydration-break advertising inventory and World Cup broadcast revenue, e.g. <https://fortune.com/2026/03/09/world-cup-hydration-breaks-tv-ads-revenue-fifa-global-viewership/> and https://www.espn.com/soccer/story/_/id/49326566/hydration-breaks-world-cup-soccer-broadcast-rights

Note on provider data: raw provider files (FIFA PMSR PDFs, SofaScore-derived arrays) are **not redistributed** in this repository. Only project-generated analytical tables are committed. See `LICENSE` and the data-availability note in `README.md`.